

Disproof of Conjecture 00000003481: Thue numbers of trees are at most 4, and $\pi(T) \leq 2 \log \Delta(T)$ fails

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Abstract

Conjecture 00000003481 claims that the Thue number of the binary tree is four, that the Thue number of general trees is unbounded, that $2 \log \Delta$ is an upper bound for the Thue number of a tree of maximum degree Δ , and that this bound is attained asymptotically by subtrees of the complete binary tree. We show that three of the four clauses are false. (i) The bound $\pi(T) \leq 2 \log \Delta(T)$ fails for every logarithm base $b > 1$. The single edge K_2 has $\Delta = 1$ and $\pi = 2 > 0 = 2 \log 1$. Even for $\Delta \geq 2$, the path P_4 has $\pi = 3 > 2 \log_b 2$ for every $b > 4^{1/3}$, including bases 2, e and 10. (ii) The Thue number of trees is *bounded*: $\pi(T) \leq 4$ for every tree (Brešar, Grytczuk, Klavžar, Niwczyk and Peterin, 2007). We include a complete proof of this bound that uses only Thue's theorem. (iii) Since $\pi \leq 4$ while $2 \log \Delta \rightarrow \infty$, the bound is never attained asymptotically. Moreover, subtrees of the complete binary tree have $\Delta \leq 3$, so they cannot form a family with $\Delta \rightarrow \infty$. The refutation of (i) is formalised in Lean 4 (core library only). So is the bound $\pi(T) \leq 4$ for every finite tree, and hence the refutation of (ii), with Thue's theorem as the only explicit hypothesis. Graphs, paths, square factors, Thue colourings, the Thue number and trees are all defined there from scratch. An independent Python script computes π for all 987 trees with at most 12 vertices. It also finds that the complete binary tree of depth 6 has $\pi = 4$, which is consistent with the remaining clause “the Thue number of the binary tree is four”.

1 The claim

The conjecture (English version): “*Definition: Thue coloring requires the color sequence of any path to contain no square factor. Conjecture: The Thue number of the binary tree is four and the Thue number of general trees is unbounded; the Thue-number upper bound for trees is twice the logarithm of the maximum degree, attained asymptotically by subtree families of the complete binary tree.*” The Chinese version has the same content.

It is a conjunction of four clauses:

- (C1) the Thue number of the binary tree is 4;
- (C2) the Thue number of general trees is unbounded;
- (C3) $\pi(T) \leq 2 \log \Delta(T)$ for every tree T , where $\Delta(T)$ is the maximum degree;
- (C4) the bound in (C3) is attained asymptotically by a family of subtrees of the complete binary tree.

We show that (C2), (C3) and (C4) are false (Theorems 2, 6, Corollaries 7, 8). One false clause already makes the conjunction false. Section 6 lists the readings that each refutation covers. Clause (C1) is not refuted: Section 5 gives what we verified about it.

2 Definitions

All graphs are simple. A *path* in a graph G is a sequence $(v_1, \dots, v_\ell)$, $\ell \geq 1$, of pairwise distinct vertices in which $v_i v_{i+1}$ is an edge for every i . Any path counts, between any two vertices; for trees these are not only root-to-leaf paths. A word s has a *square factor* if $s = axxb$ for words a, x, b with x nonempty. A colouring c of $V(G)$ is a *Thue colouring* (also called *nonrepetitive*) if no path has a colour sequence $c(v_1) \cdots c(v_\ell)$ with a square factor. The *Thue number* $\pi(G)$ is the least number of colours of a Thue colouring [2]. A Thue colouring is in particular proper (take x of length 1), and π is monotone under taking subgraphs. A word is *square-free* if it has no square factor.

3 Clause (C3) is false: K_2 and P_4

For a base $b > 1$ and integers $k \geq 0$, $\Delta \geq 1$,

$$k \leq 2 \log_b \Delta \iff b^k \leq \Delta^2, \quad (1)$$

because $t \mapsto b^t$ is increasing. For a rational base $b = p/q$ ($p > q \geq 1$) the right-hand side reads $p^k \leq \Delta^2 q^k$. A larger base makes the bound smaller. So the claim for a base b implies the claim for every base $b' \in (1, b]$. Every real $b > 1$ lies above some rational $p/q > 1$, so refuting all rational bases $p/q > 1$ refutes all real bases $b > 1$.

Lemma 1. $\pi(K_2) = 2$ and $\pi(P_4) = 3$.

Proof. K_2 : one colour gives the colour sequence aa on the path through the edge. Two distinct colours work, because the only paths are single vertices and the edge.

$P_4 = v_1 v_2 v_3 v_4$: suppose there is a Thue colouring with two colours. Then $c(v_1) \neq c(v_2) \neq c(v_3) \neq c(v_4)$, because otherwise one edge carries the square aa . With two colours this forces $c(v_1) = c(v_3)$ and $c(v_2) = c(v_4)$, so the whole path has colour sequence $abab$, a square. Conversely, the colouring $1, 2, 3, 1$ is a Thue colouring. Every path of P_4 is a factor of 1231 or of its reverse. Their factors of length 2 have distinct letters, and the only factor of length 4 is 1231 (or 1321), which is not of the form xx . $\square$

Theorem 2. For every base $b > 1$, the inequality $\pi(T) \leq 2 \log_b \Delta(T)$ fails for some tree:

- $T = K_2$: $\Delta = 1$ and $\pi = 2 > 0 = 2 \log_b 1$, for every $b > 1$;
- $T = P_4$: $\Delta = 2$ and $\pi = 3 > 2 \log_b 2$ for every $b > 4^{1/3} \approx 1.587$; for example, $2 \log_2 2 = 2$, $2 \ln 2 \approx 1.386$ and $2 \log_{10} 2 \approx 0.602$;
- $T =$ the complete binary tree B_6 of depth 6 (computer search, Section 5): $\Delta = 3$ and $\pi = 4 > 2 \log_b 3$ for every $b > \sqrt{3}$; for example, $2 \log_2 3 \approx 3.17$.

Proof. Use Lemma 1 and (1). $b^2 \leq 1$ is false for $b > 1$, and $b^3 \leq 4$ is false for $b > 4^{1/3}$. $\square$

The last item shows that even the family named in (C4) violates the bound in (C3). The bound also fails when it is rounded up: $\lceil 2 \log_b 1 \rceil = 0 < 2$, and $\lceil 2 \log_2 2 \rceil = \lceil 2 \ln 2 \rceil = 2 < 3$. It fails as well for the Thue *choice* number $\pi_{ch} \geq \pi$.

4 Every tree has $\pi(T) \leq 4$

This is a theorem of Brešar, Grytczuk, Klavžar, Niwczyk and Peterin [3]; see also Wood's survey [7, Thm. 4.1]. For completeness we give a full proof. It uses the following classical theorem.

Theorem 3 (Thue 1906 [1]). *There is an infinite square-free word $w = w_0w_1w_2\cdots$ over $\{1, 2, 3\}$.*

Lemma 4 (Kündgen–Pelsmajer [4]; [7, Lemma 3.3]). *Let w be a square-free word over $\{1, 2, 3\}$. Let u be the word obtained by inserting the letter 4 after every second letter, $u = w_0w_14w_2w_34w_4w_54\cdots$. Then (i) any three consecutive letters of u are pairwise distinct, and (ii) u is square-free.*

Proof. (i) Three consecutive letters of u have one of the forms $w_{2j}w_{2j+1}4$, $w_{2j+1}4w_{2j+2}$ or $4w_{2j+2}w_{2j+3}$. The two letters of w in each are consecutive in w , so they differ, since w has no factor aa . Neither of them equals 4.

(ii) Let xx be a factor of u . Delete all letters 4 from it. The result is a factor of w of the form $x'x'$, where x' is x with its letters 4 deleted. This is because the two copies of x have their 4's in the same positions. If x' is empty, then x consists of 4's only, so u has the factor 44, which contradicts (i). Otherwise $x'x'$ is a square in w , which is a contradiction. $\square$

Lemma 5 (folding). *Let u be a square-free word in which any three consecutive letters are pairwise distinct. Let $m, d_1, d_2 \geq 0$ with $m + \max(d_1, d_2) < |u|$ (or u infinite). Then the word*

$$s = u_{m+d_1}u_{m+d_1-1}\cdots u_{m+1}u_mu_{m+1}\cdots u_{m+d_2}$$

is square-free.

Proof. Index s by $t \in \{-d_1, \dots, d_2\}$, so that $s_t = u_{m+|t|}$. Suppose s has a square factor occupying the positions $i, \dots, i+2k-1$ ($k \geq 1$). Then $s_t = s_{t+k}$ for $i \leq t \leq i+k-1$.

If $i \geq 0$, the factor lies in the part $t \geq 0$, where $s_t = u_{m+t}$. So it is a square factor of u , a contradiction. If $i+2k-1 \leq 0$, it lies in the part $t \leq 0$, which is the reverse of a factor of u . The reverse of a square is a square, so this is again a contradiction. So $i < 0 < i+2k-1$.

Reversing s , i.e. replacing t by $-t$ and swapping d_1 and d_2 , maps squares to squares and fixes the position 0. If 0 lies in the second half $[i+k, i+2k-1]$ of the square, then after reversal it lies in the first half. So we may assume $i < 0 \leq i+k-1$, i.e. $a := -i \in [1, k-1]$.

Case $a \leq k-2$. The positions $-1, 0, 1$ lie in the first half $[i, i+k-1]$. Hence $s_{k-1} = s_{-1} = u_{m+1}$ and $s_{k+1} = s_1 = u_{m+1}$. Since $k-1 \geq 1 > 0$, we have $s_{k-1} = u_{m+k-1}$ and $s_{k+1} = u_{m+k+1}$. So $u_{m+k-1} = u_{m+k+1}$, which contradicts the hypothesis on three consecutive letters.

Case $a = k-1$ (so $k \geq 2$). The square occupies $[1-k, k]$, and for $t \in [1-k, 0]$ we have $u_{m-t} = s_t = s_{t+k} = u_{m+t+k}$. Put $j = -t \in [0, k-1]$ and $y_j = u_{m+j}$ for $0 \leq j \leq k$. Then $y_j = y_{k-j}$, so $y_0 \cdots y_k$ is a palindrome of length $k+1 \geq 3$. If k is even, then $y_{k/2-1} = y_{k/2+1}$. If k is odd, then $y_{(k-1)/2} = y_{(k+1)/2}$. Both contradict the hypothesis. $\square$

Theorem 6 ([3]). *Every tree T , finite or infinite, satisfies $\pi(T) \leq 4$.*

Proof. Let u be the infinite word of Lemma 4 built from Thue's word (Theorem 3). Fix a root r , and colour every vertex v with $u_{\text{dist}(r,v)}$. Every non-root vertex has exactly one neighbour of smaller depth, its parent, and every edge joins a vertex to its parent. Let $P = (v_0, \dots, v_\ell)$ be a path. An interior vertex v_j cannot have both v_{j-1} and v_{j+1} as its parent, because they are distinct. So the depth sequence along P has no interior local maximum. It changes by exactly 1 at each step, so it first decreases by steps of 1 down to the depth m of the top vertex, and then increases by steps of 1. Hence the colour sequence of P is exactly the word s of Lemma 5, with d_1, d_2 the lengths of the two monotone parts. That word is square-free. $\square$

For a finite tree only a finite prefix of u is needed. For finite trees, this proof is formalised in Lean, with Thue’s theorem as an explicit hypothesis (Section 7). Its existence for every length is still Thue’s theorem. Brešar et al. [3] also show that $\pi(T) = 4$ for some trees, so the bound is tight. Section 5 gives an explicit example.

Corollary 7. *Clause (C2) is false: $\sup_T \pi(T) = 4 < \infty$ over all trees.*

Corollary 8. *Clause (C4) is false. Let (T_n) be any family of trees with $\Delta(T_n) \rightarrow \infty$. Then $\pi(T_n)/(2 \log \Delta(T_n)) \leq 4/(2 \log \Delta(T_n)) \rightarrow 0$, so the bound $2 \log \Delta$ is not attained asymptotically, in any base. Moreover, a subtree of the complete binary tree has maximum degree at most 3. So subtree families of the complete binary tree have bounded Δ , and an asymptotic statement in Δ about them is impossible. Every such subtree satisfies $\pi \leq 4$, while $2 \log_b \Delta \leq 2 \log_b 3$ is fixed.*

5 What we know about clause (C1)

Let B_h be the complete binary tree of depth h . Its root has degree 2, and its other internal vertices have degree 3. An exhaustive backtracking search (`verify.py`) gives

$$\pi(B_1) = 2, \quad \pi(B_2) = \pi(B_3) = \pi(B_4) = \pi(B_5) = 3, \quad \pi(B_6) = 4.$$

The colourings found for $h \leq 5$ are re-checked on all paths. The non-existence of a Thue 3-colouring of B_6 (127 vertices) is confirmed by two different searches: breadth-first order with colour-symmetry breaking, and depth-first order without it. By monotonicity and Theorem 6, $\pi(B_h) = 4$ for all $h \geq 6$, and also for the infinite binary tree. So (C1) holds for the infinite or sufficiently deep complete binary tree, and we do not refute it. Under the reading “every binary tree has Thue number 4” it would be false, because P_4 is a binary tree with $\pi = 3$. We do not rely on this reading. The result $\pi(B_6) = 4$ is a computer result and is not formalised in Lean.

6 Readings covered

The Lean result `conjecture_00000003481_false` refutes the literal (C3). Under the asymptotic reading of (C3), the disproof goes through (C2), whose refutation, $\pi(T) \leq 4$ for every tree, is proved completely in Section 4. That refutation is also formalised in Lean, modulo Thue’s theorem as an explicit hypothesis (Section 7).

- **(C3) as a universal bound** $\pi(T) \leq 2 \log_b \Delta(T)$ (the literal reading). False for every base $b > 1$ by K_2 . False for every $b > 4^{1/3}$, in particular $b = 2, e, 10$, by P_4 even when $\Delta \geq 2$ is required, and false for $b > \sqrt{3}$ by B_6 . Also false with $\lceil \cdot \rceil$, and for the choice number $\pi_{ch} \geq \pi$.
- **(C3) only “for large Δ ” or up to $O(1)$.** Then (C3) is true but trivial, by Theorem 6: $4 \leq 2 \log_b \Delta$ as soon as $\Delta \geq b^2$. In this case (C4) is false (Corollary 8), and so is (C2). The conjunction is false under every reading.
- **(C2) is false under every reading of “Thue number”** as the chromatic-type parameter defined in the statement (Corollary 7), for finite and for infinite trees. *Disclosure:* for the different parameter π_{ch} (nonrepetitive *list* colouring), trees do have unbounded values (Fiorenzi, Ochem, Ossona de Mendez and Zhu [5]). The statement defines colourings, not list colourings, so this does not apply. Even for π_{ch} , the literal (C3) fails by K_2 and P_4 .

7 Formal verification (Lean 4)

The directory `lean4/` contains a Lean 4.19.0 project that uses only the core library. `Main.lean` defines, from scratch:

- `Graph` (vertices $0, \dots, n - 1$, Boolean adjacency);
- `IsPath` (nonempty, duplicate-free, consecutive vertices adjacent; all paths, not only root-to-leaf ones);
- `HasSquare s` ($s = a x x b$ with $x \neq []$);
- `Nonrepetitive, Colorable G k` and `IsThueNumber G k` (the least such k , proved unique);
- `IsTree`: a nonempty simple graph in which any two vertices are joined by exactly one path (the textbook definition; for K_2 and P_4 we also check “connected with $n - 1$ edges”), and `maxDeg`;
- `LogBound p q Δ k` $:= p^k \leq \Delta^2 q^k$, which is (1) for the base p/q .

The main results are:

- `thue_K2`: $\pi(K_2) = 2$, and `thue_P4`: $\pi(P_4) = 3$;
- `K2_tree` and `P4_tree`: both are trees, with `maxDeg` 1 and 2;
- `conjecture_00000003481_false`: for all $p > q$, $\neg \text{Claim } p \ q$, where `Claim p q` says that every tree with $\Delta \geq 1$ and Thue number k satisfies `LogBound p q Δ k`;
- `claimCol_false`: the same for the colouring form “some Thue colouring with $\leq 2 \log \Delta$ colours exists”, which does not use the minimum;
- `claimDeg2_false`: the claim restricted to $\Delta \geq 2$ is false whenever $p^3 > 4q^3$, for example for bases 2 and $27/10 < e$;
- `claim_mono`: a larger base gives a stronger claim;
- `nonvacuous`: the hypotheses are satisfiable, and Thue colouring is a nontrivial condition.

Clause (C2) in Lean (Section 7 of `Main.lean`). The hypothesis is

`ThueHyp` $:= \forall L, \exists w : \mathbb{N} \rightarrow \mathbb{N}, (\forall i, w_i < 3) \wedge \forall i k, 0 < k \rightarrow i + 2k \leq L \rightarrow \exists j < k, w_{i+j} \neq w_{i+j+k}$,

i.e. Thue’s theorem [1, 6] in finite form: a square-free ternary word of every length. The list form `ThueHypList` (lists of every length with entries < 3 and $\neg \text{HasSquare}$) implies it (`thueHyp_of_list`). `thueHyp_40` checks the hypothesis for $L = 40$ by `decide`. From it, Lean proves:

- `kp_sqfree, kp_ne1, kp_ne2`: Lemma 4 for $u_i = 3$ if $i \equiv 2 \pmod{3}$ and $u_i = w_{2\lfloor i/3 \rfloor + (i \bmod 3)}$ otherwise;
- `fold_sqfree`: Lemma 5, for $t \mapsto u_{m+|t-a|}$, with all six positions of the turning point relative to the square;
- `valley`: along any path of a graph with a layering with unique parents, the depth is $m + |t - a|$;

- `tree_layering`: every tree (in the sense of `IsTree`) has such a layering, namely the depth $d(v) = |P_v| - 1$, where P_v is the unique path from 0 to v ;
- `tree_colorable_four`: assuming `ThueHyp`, every tree has a Thue 4-colouring; hence $\pi(T) \leq 4$ (`thueNumber_le_four`);
- `conjecture_C2_false`: assuming `ThueHyp`, it is not the case that for every k some tree T satisfies $\neg \text{Colorable } T \ k$. The variant with π is `conjecture_C2_false'`; the variant assuming the list form is `conjecture_C2_false_list`.

These use the same `Graph`, `IsPath`, `HasSquare`, `Colorable` and `IsTree` as the main theorem. Nonrepetitiveness of the explicit colourings is decided over all candidate vertex sequences. The completeness of this enumeration is proved: by a pigeonhole lemma, a path has at most n vertices. Only the axioms `propext`, `Classical.choice` and `Quot.sound` are used. There is no `sorry` and no `native_decide`.

The following are *not* in Lean, only here and in `verify.py`:

- Thue's theorem itself (Theorem 3), which is an explicit hypothesis;
- the case of infinite trees in Theorem 6 (Lean graphs are finite);
- Corollary 8;
- the computation $\pi(B_6) = 4$;
- the passage from rational to real bases, which is the one-line monotonicity argument after (1); its rational form is `claim_mono`.

8 Independent check (Python)

`verify.py` uses only the standard library. It does the following:

- computes $\pi(K_2) = 2$ and $\pi(P_4) = 3$ by enumerating all colourings and all paths;
- generates all trees with $n \leq 12$ vertices up to isomorphism (987 trees; the counts match OEIS A000055) and computes π for each by backtracking, re-checking each colouring on all paths;
- finds the maximum to be 3 for $n \leq 12$, and counts the trees that violate (C3): 10 for base 2 and 661 for base e ;
- computes $\pi(B_h)$ for $h \leq 6$, with the two independent searches for B_6 . Both searches are fast: each shows that B_6 (127 vertices) has no Thue 3-colouring in well under a second (about 0.03s). The whole script runs in about 7s;
- checks the construction of Theorem 6. It builds a ternary square-free word, applies the Kündgen–Pelsmajer insertion and checks the depth colouring on all 11006 rooted trees with $n \leq 12$ and on $B_1, \dots, B_8$.

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